AI-Native Transceiver Design for Next-Generation Wireless Networks

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1 Background and Motivation

A conventional transceiver is a chain of separately designed blocks — each optimal alone, jointly only as good as the assumptions linking them. Deep learning replaces the chain with an **autoencoder** [1], or estimation and detection with a **learned receiver** [2].

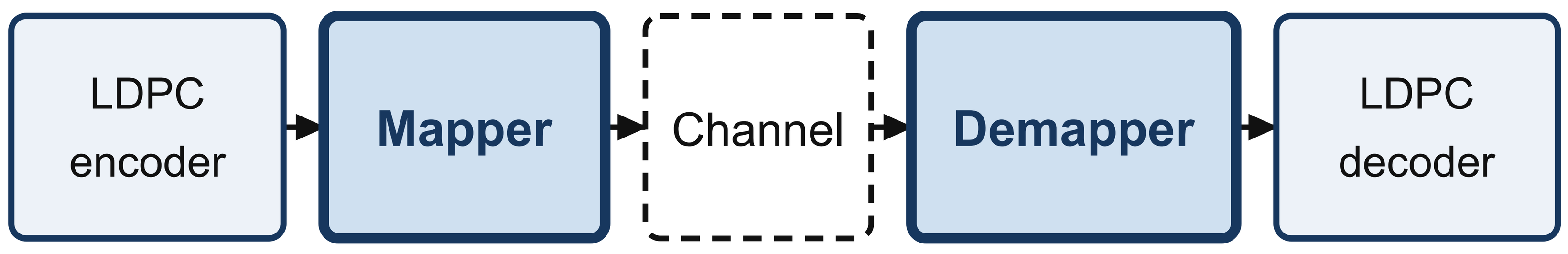
- Training needs a **differentiable channel model** — a real channel is a black box.
- Most reported gains are on **AWGN** alone, where the classical receiver is already optimal.

Two questions follow, settled here by measurement: **how much is end-to-end learning worth** against a classical receiver doing its best, and **can it be trained without a channel model at all?**

2 Objectives and Contributions

- Measure the gain of a learned transceiver against a **matched** classical baseline, on AWGN and on Rayleigh block fading with **pilot-estimated CSI**.
- Train the transmitter with **no channel model at all** — reinforcement learning in place of backpropagation through the channel — and measure what that costs.
- Keep it honest: the transmitter **never sees the channel**; pilots are charged for (rate **R(L−P)/L**) and so is estimation error (**$N_0(1+1/P)/|\hat{h}|^2$**) — omitting that factor understates the noise by **1.76 dB** at P = 2.
- Diagnose **by intervention, not by argument**, why AWGN-trained weights fail on a fading channel.

3 System Model

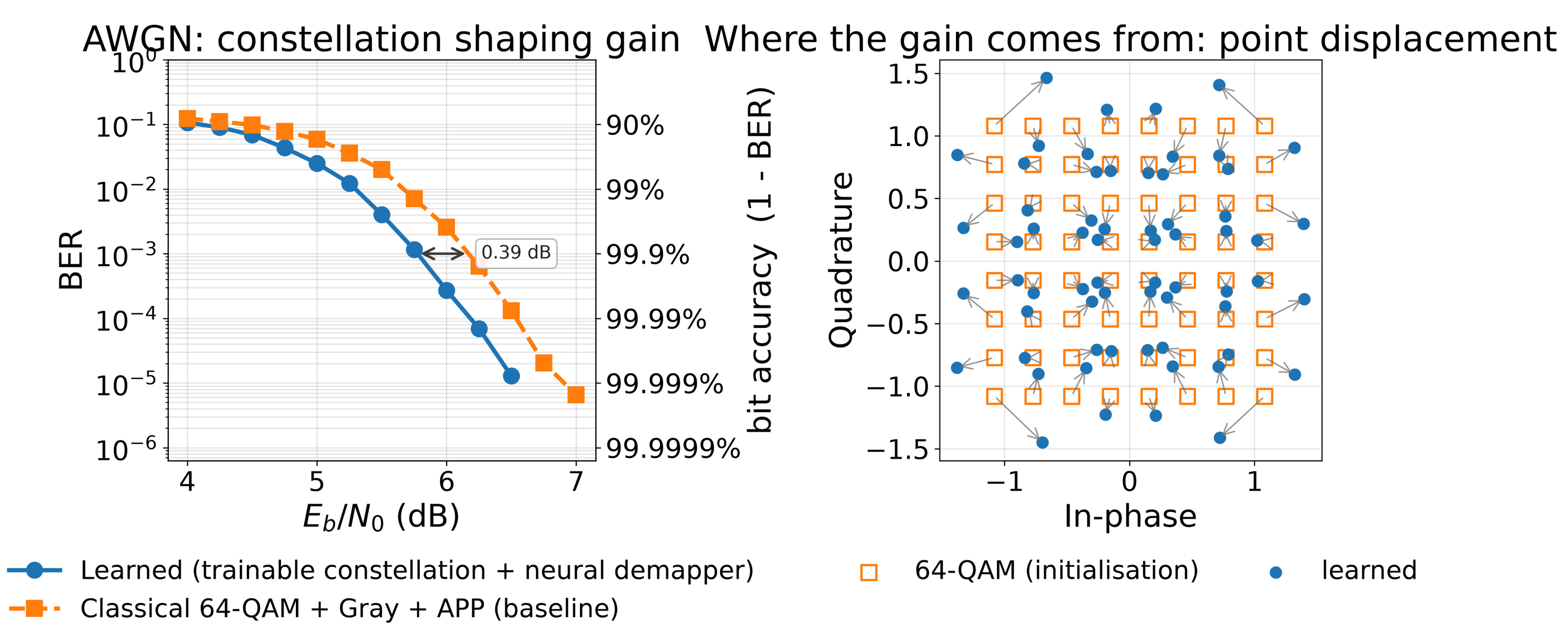


Learned	trainable constellation	neural network
Classical	64-QAM, Gray	APP demapper

Channel: AWGN | Rayleigh block fading, L = 25, P pilot symbols, LS estimate

Only the two shaded blocks differ. The LDPC code, the channel call and the seeds are **shared**, so any BER difference is down to the mapper and demapper alone.

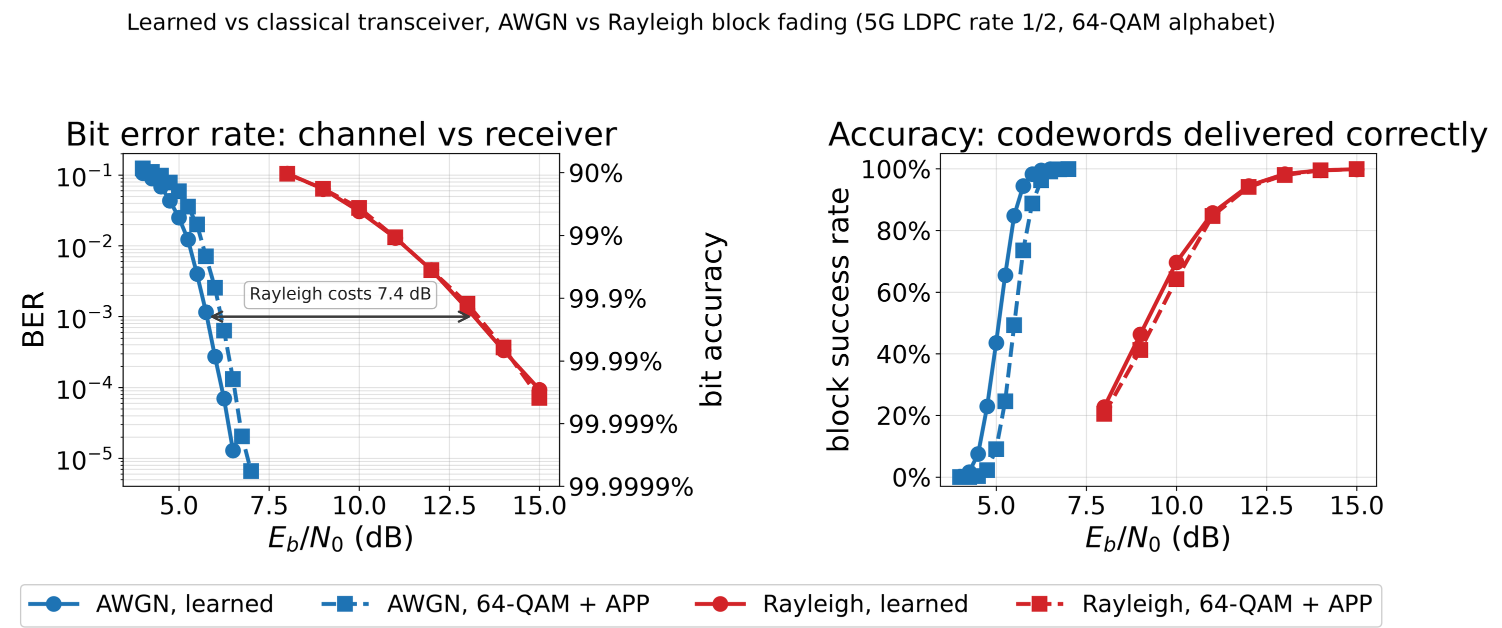
4 Where the AWGN Gain Comes From



- Under AWGN, APP demapping is **already optimal** — the receiver cannot be the source of the gain.
- All of it comes from moving the transmitted points: the learned constellation pulls off the rigid lattice, spending more power on the most **confusable** symbols.

5 Performance Analysis

Both channels, both receivers, one axis — colour is the channel, line style the receiver.



- Fading costs **5.9–8.8 dB**; the choice of receiver moves things by under **0.4 dB**.
- **AWGN:** the learned system wins by **+0.37 / +0.39 / +0.35 dB** at BER 1e-2 / 1e-3 / 1e-4. APP demapping is already optimal here, so this is **constellation shaping alone**.
- **Rayleigh with estimated CSI: no measurable gain** (+0.01 / +0.11 / −0.15 dB) — the 1e-4 figure **changed sign** between two runs of the same systems, so it is noise, not a difference.

6 Why AWGN Weights Fail on Fading

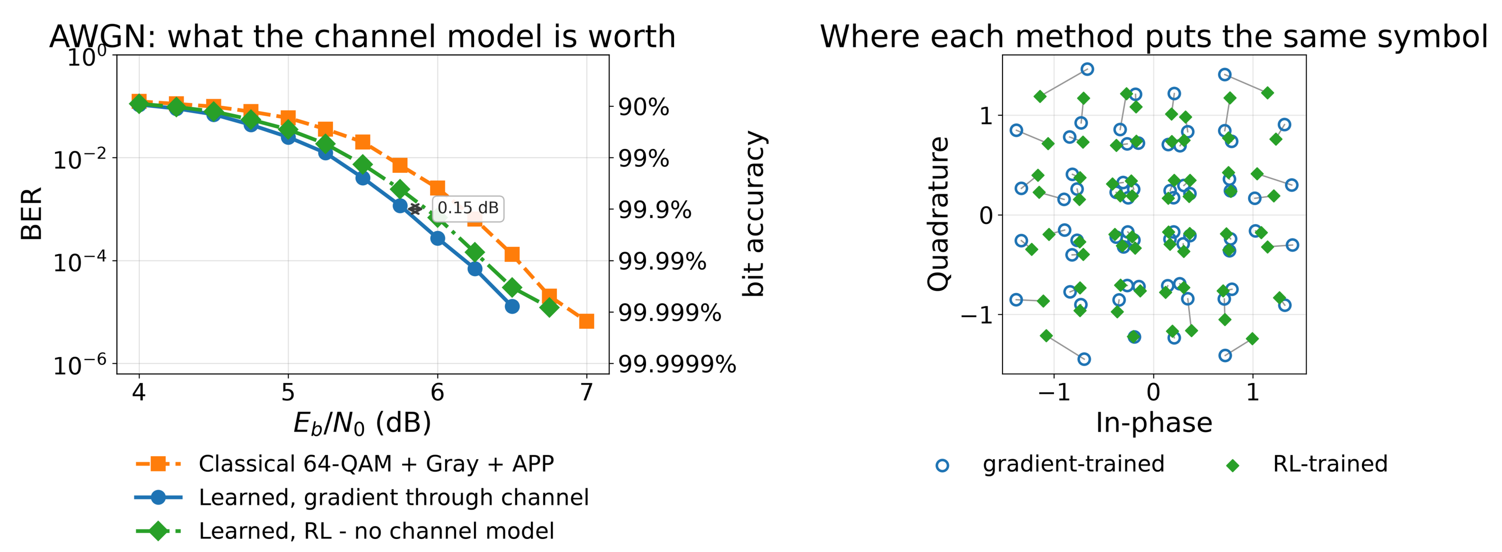
Reused on fading, AWGN-trained weights give a BER curve that **rises** with SNR. The weights were held fixed and **one thing changed at a time**:

- Clamping the network's outputs: under 4% change — **the rise remains**.
- Erasing deep-fade symbols: **worse** above 10 dB.
- Clamping the noise level **reported to the demapper** back into its training range: **the curve becomes physical**, with no weight changed.

The failure is on the **input** side, in the **quiet** tail: at 18 dB **89%** of symbols are quieter than anything seen in training, against 4.6% in deep fades. Retraining restores a monotonic curve — **1e-3** at 13.2 dB, **1e-4** at 14.9 dB.

7 Learning Without a Channel Model

Every result above needed a differentiable channel to backpropagate through; a deployed transmitter has none. Here the receiver trains normally — its gradient never crosses the channel — while the transmitter perturbs its own symbols and updates from **one scalar per codeword** [4]. **The channel is sampled, never differentiated.**



- **Price of dispensing with the channel model: +0.12 / +0.15 / +0.12 dB** at BER 1e-2 / 1e-3 / 1e-4.
- The RL-trained system still beats classical 64-QAM by **0.24 dB** — with **no channel model anywhere in training**.
- Both methods put the same symbol in nearly the same place (right panel), so RL is finding the gradient method's solution rather than a different one.

8 Conclusion and Next Steps

End-to-end learning buys **0.37 dB on AWGN** and **nothing once the channel fades and CSI is estimated**. So accuracy is not the case for AI-native transceivers — the case is that the design **needs no channel model**, and that now carries a **measured price of 0.12–0.15 dB** rather than an assumption.

- **RQ1 — no channel model. Answered.** RL training costs 0.12–0.15 dB against gradient training, and still beats the classical receiver by 0.24 dB.
- **RQ2 — fast adaptation.** Remaining work. Meta-learning [3] targets exactly the failure measured above — motivation **measured here, not cited.**

64,000 codewords per SNR point; points below 30 block errors dropped, not plotted. Seeds fixed throughout, and every number here is measured in the notebook rather than quoted. Sionna [5] + PyTorch. **Scope:** flat block fading only — no multipath, Doppler or OFDM.

[1] T. O'Shea, J. Hoydis, "Deep Learning for the Physical Layer," arXiv:1702.00832, 2017.
[3] S. Smeka J. et al., "Enhanced Reptile for Rapid Adaptation," SCSE, IEEE, 2025.
[5] NVIDIA Sionna, autoencoder tutorial notebook — the starting point this work extends.
[2] H. Ye, G. Y. Li, B.-H. F. Juang, "Deep Learning for Channel Estimation and Detection in OFDM," arXiv:1708.08514, 2017.
[4] N. C. Luong et al., "Deep RL in Communications and Networking: A Survey," IEEE Commun. Surveys Tuts. 21(4), 2019.